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## Supplemental information

# The genome of African manatee *Trichechus senegalensis* reveals secondary adaptation to the aquatic environment

**Xin Huang, Guixin Dong, Huizhong Fan, Wenliang Zhou, Guangping Huang, Dengfeng Guan, Delu Zhang, and Fuwen Wei**

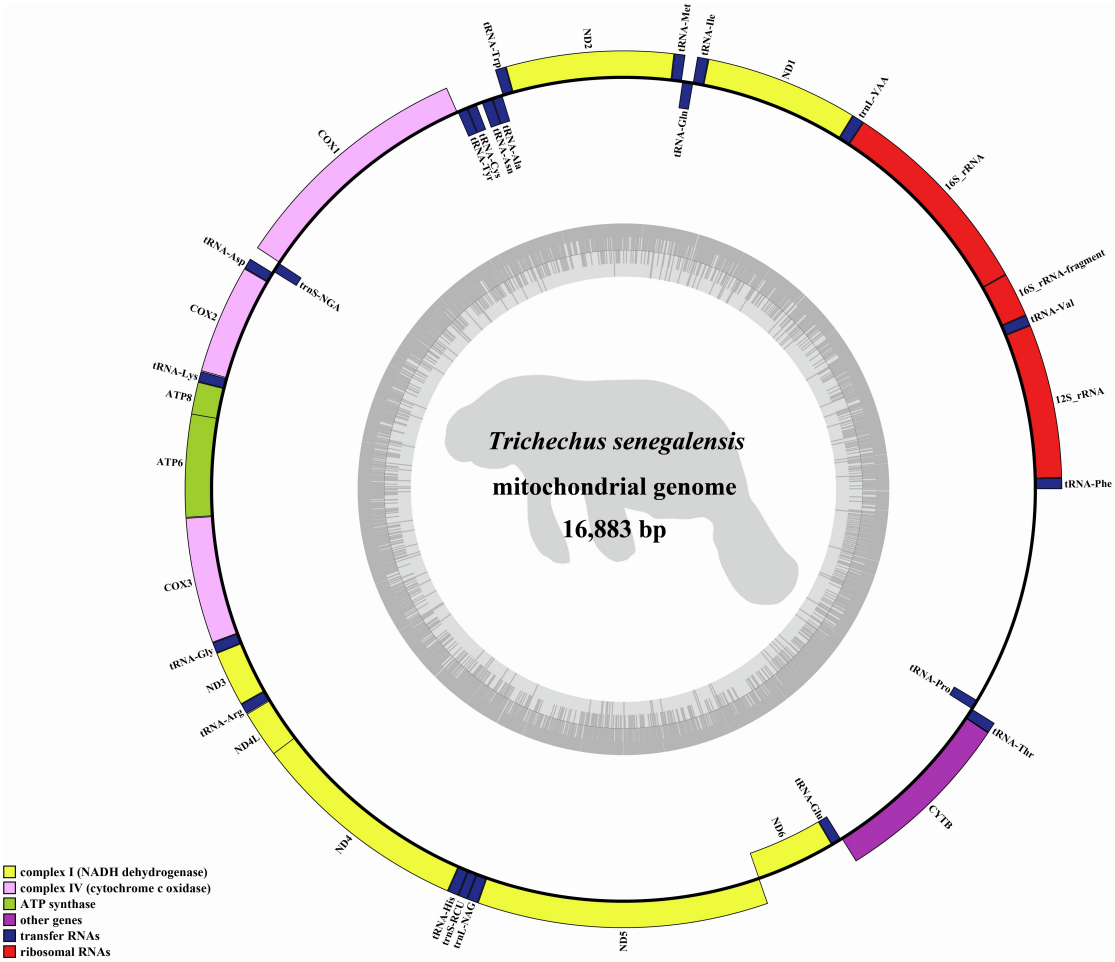


**Figure S1. Scheme of the circular mitochondrial genome of the African manatee, related to STAR Methods.**

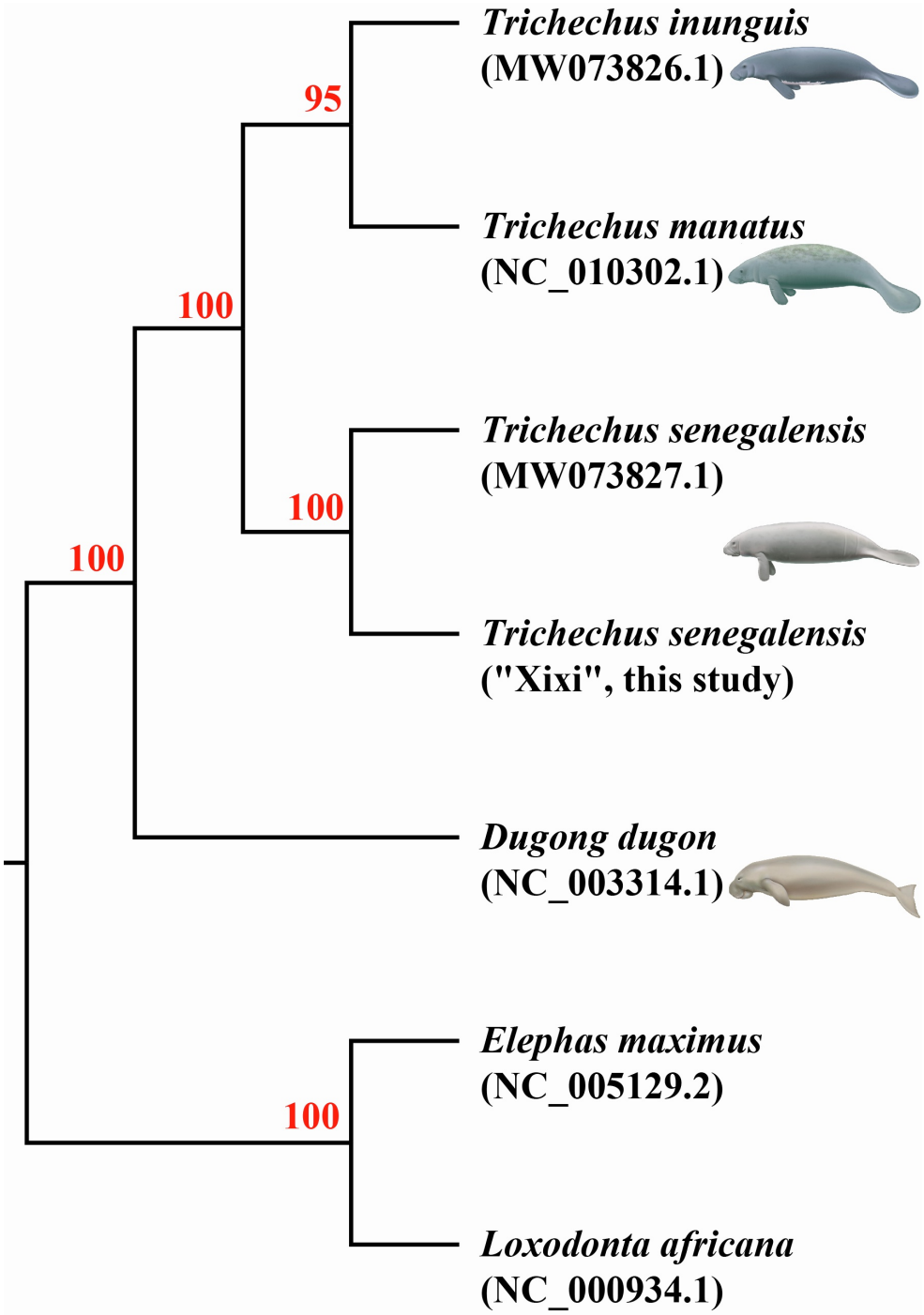


**Figure S2. Phylogeny reconstruction of the order Sirenia at the mitochondrial level, related to STAR Methods**. The ML tree was reconstructed with RAxML based on the concatenated 13 mitochondrial protein-coding genes. The bootstrap values are shown in red color in each node. The specimen used for genome sequencing was clustered together with the African manatee with high support (bootstrap value = 100).

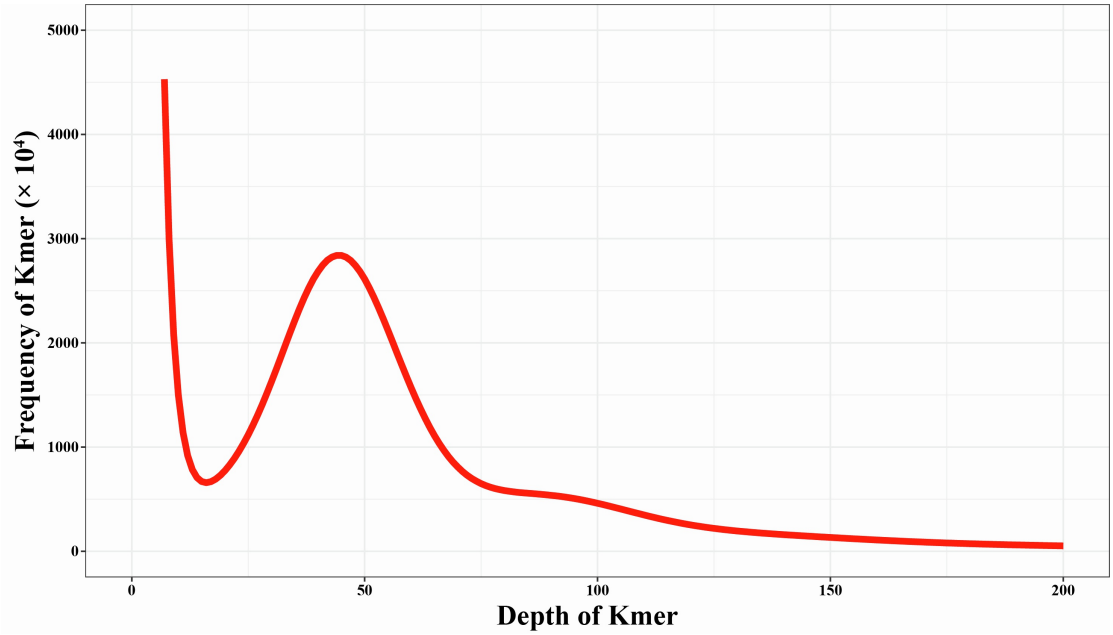


**Figure S3. K-mer (k = 17) distribution of the African manatee genome, related to Figure 1.**

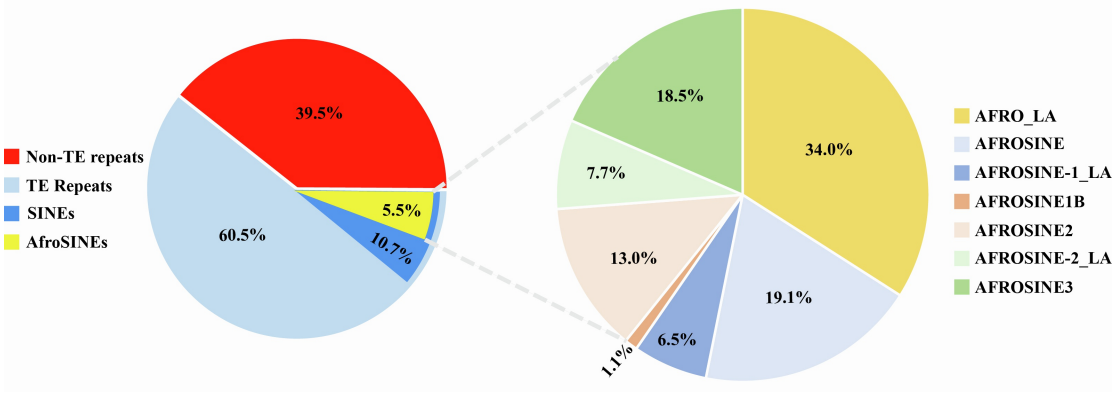


**Figure S4. The percentage of the AfroSINEs identified in the African manatee genome, related to Figure 1.** Different repeated types in AfroSINEs were shown on the right.

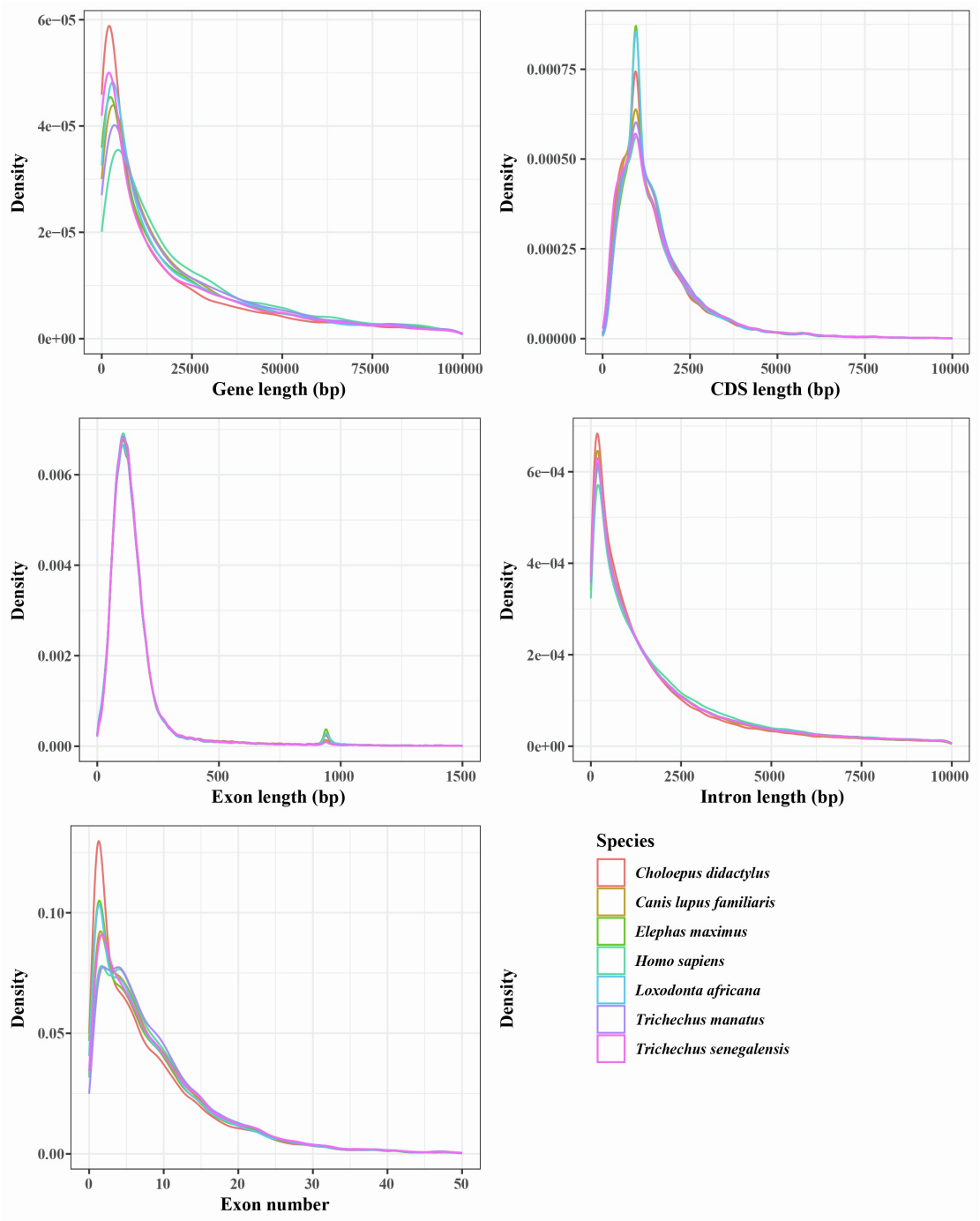


**Figure S5. The distribution of the gene length, the CDS length, the exon length, the intron length, and the exon number of each transcript, related to Figure 1.**

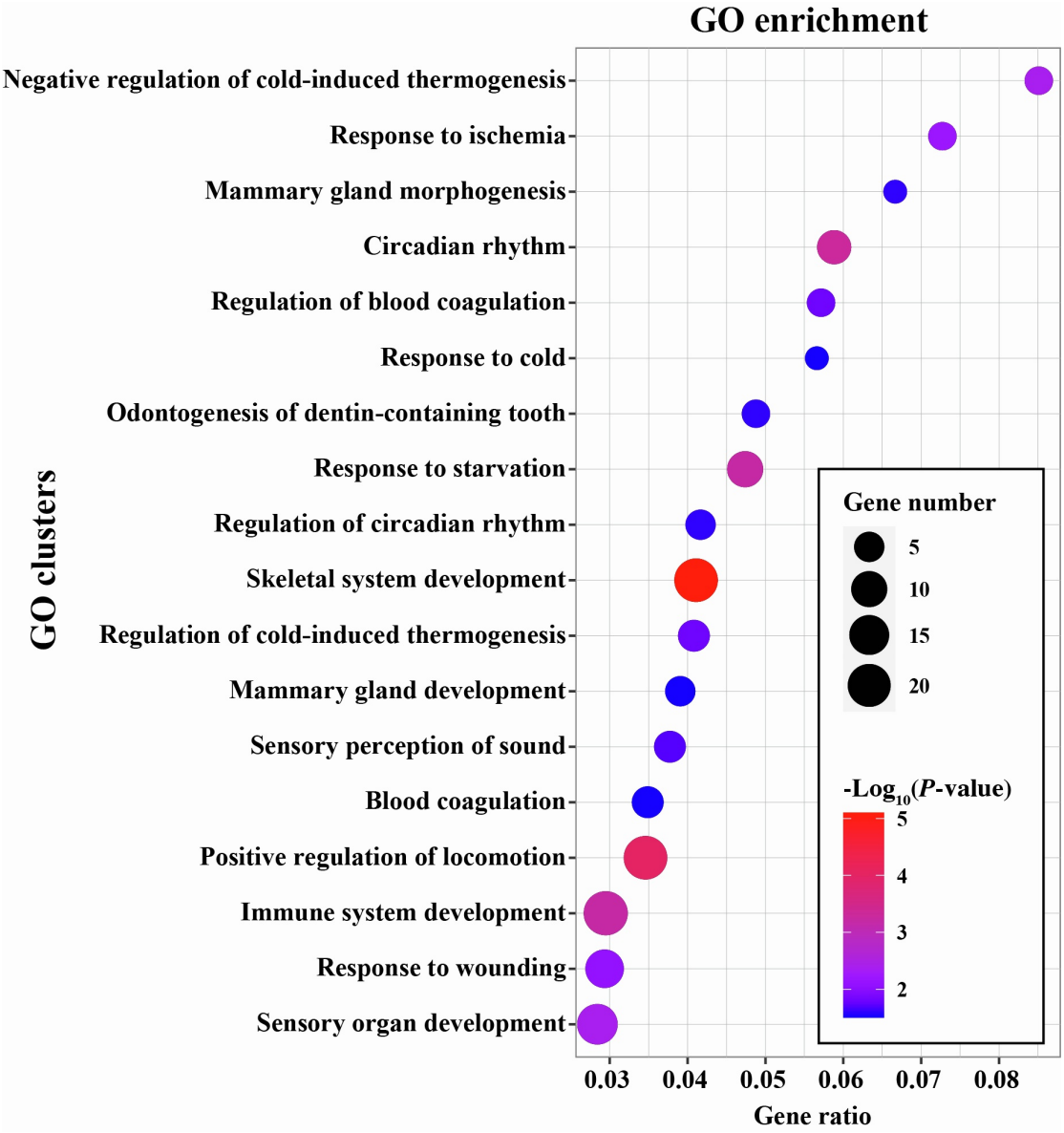


**Figure S6. The significantly enriched GO terms for positively selected genes in the African manatee genome, related to Figure 2.** Visualization as Figure 2C.

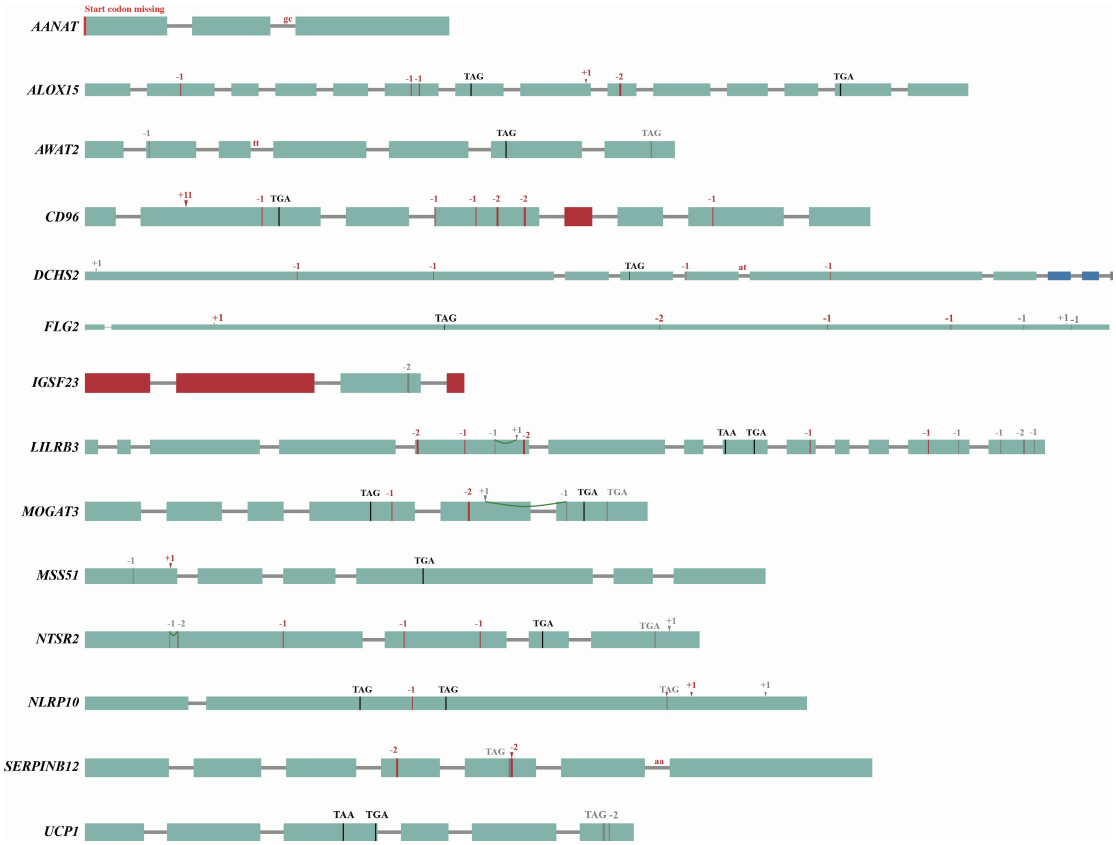


**Figure S7. Inactivating mutations in gene *AANAT*, *ALOX15*, *AWAT2*, *CD96*, *DCHS2*, *FLG2*, *IGSF23*, *LILRB3*, *MOGAT3*, *MSS51*, *NTSR2*, *NLRP10*, *SERPINB12*, and *UCP1* of the African manatee, related to Figure 2.** Visualization as Figure 2E.

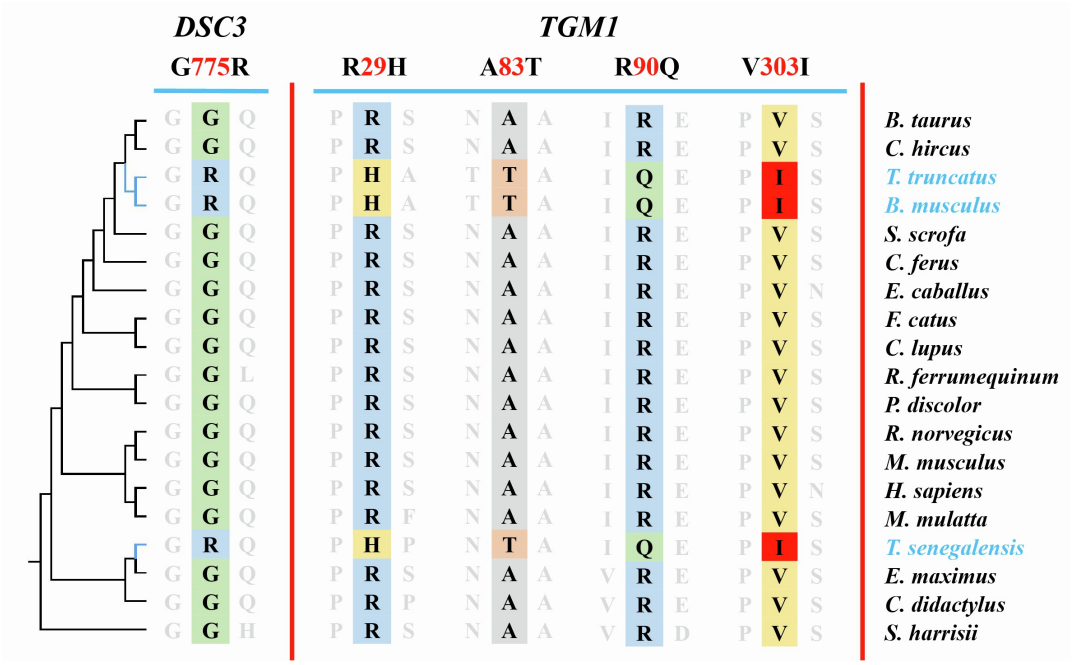


**Figure S8. The convergent amino acid substitutions in genes *DSC3* and *TGM1* of fully aquatic mammals, related to Figure 3.**

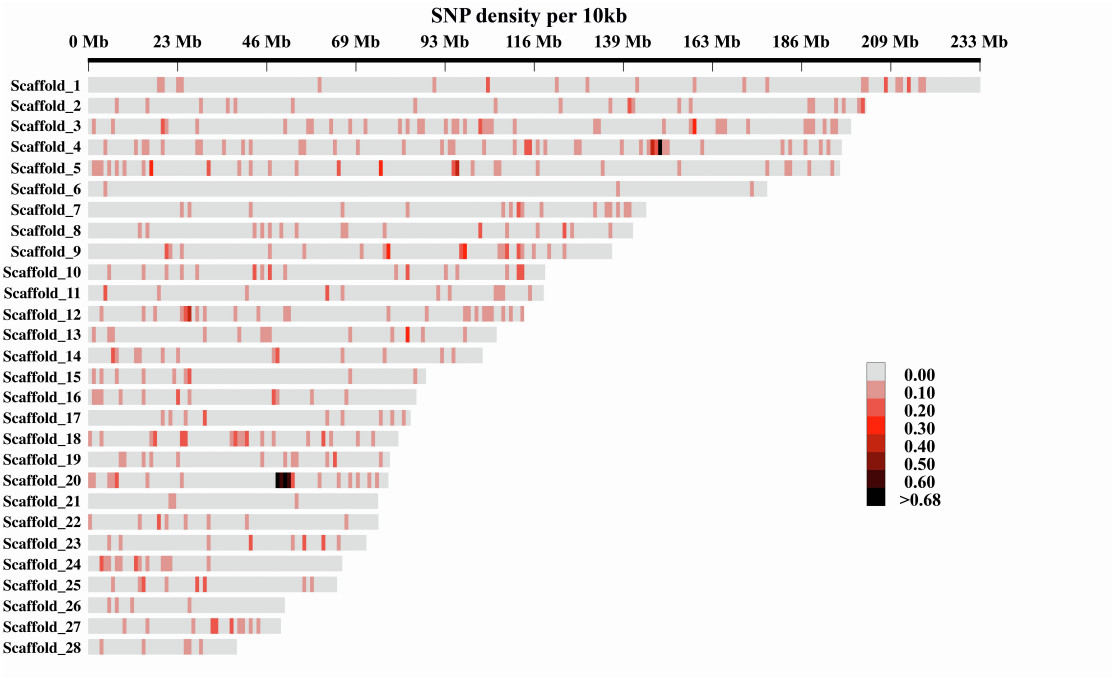


**Figure S9. The density of SNPs per 10 kb across chromosome-length scaffolds of the African manatee, related to Figure 4.**

**Table S1. Summary of whole genome sequenced reads, related to Table 1.**

| Data type | Cell/Pair | Sequence number | Sum of lengths (bp) | Minimum length (bp) | Average length (bp) | Maximum length (bp) |
|---|---|---|---|---|---|---|
| HiFi | m64181_220520_085259 | 1,917,306 | 31,925,492,205 | 655 | 16,651.20 | 67,560 |
| | m64181_220521_194732 | 1,836,456 | 30,599,647,460 | 369 | 16,662.30 | 63,080 |
| | m64181_220523_064416 | 1,583,416 | 25,898,052,414 | 573 | 16,355.80 | 67,980 |
| | m64446e_220513_084159 | 1,139,547 | 19,204,851,524 | 631 | 16,853.10 | 63,775 |
| | m64446e_220514_193602 | 894,331 | 14,874,010,224 | 680 | 16,631.40 | 57,381 |
| | m64446e_220517_091820 | 1,205,217 | 19,395,897,152 | 182 | 16,093.30 | 61,988 |
| Hi-C | R1 | 1,231,225,770 | 184,683,865,500 | 150 | 150 | 150 |
| | R2 | 1,231,225,770 | 184,683,865,500 | 150 | 150 | 150 |
| Illumina | R1 | 543,669,683 | 81,550,452,450 | 150 | 150 | 150 |
| | R2 | 543,669,683 | 81,550,452,450 | 150 | 150 | 150 |

**Table S2. Length of the 28 pseudochromosomes of the African manatee genome, related to Figure 1.**

| Type | Scaffold name | Length (bp) | Pseudochromosome name |
|---|---|---|---|
| Pseudochromosome | Scaffold_1 | 233,018,420 | Chr1 |
| | Scaffold_2 | 203,250,200 | Chr2 |
| | Scaffold_3 | 199,232,634 | Chr3 |
| | Scaffold_4 | 196,794,457 | Chr4 |
| | Scaffold_5 | 196,358,007 | Chr5 |
| | Scaffold_6 | 177,308,397 | ChrX |
| | Scaffold_7 | 145,687,139 | Chr6 |
| | Scaffold_8 | 142,222,000 | Chr7 |
| | Scaffold_9 | 136,758,213 | Chr8 |
| | Scaffold_10 | 119,264,650 | Chr9 |
| | Scaffold_11 | 118,899,908 | Chr10 |
| | Scaffold_12 | 113,759,247 | Chr11 |
| | Scaffold_13 | 106,591,697 | Chr12 |
| | Scaffold_14 | 102,939,642 | Chr13 |
| | Scaffold_15 | 88,142,921 | Chr14 |
| | Scaffold_16 | 85,612,991 | Chr15 |
| | Scaffold_17 | 84,110,478 | Chr16 |
| | Scaffold_18 | 80,919,913 | Chr17 |
| | Scaffold_19 | 78,697,699 | Chr18 |
| | Scaffold_20 | 78,268,128 | Chr19 |
| | Scaffold_21 | 75,604,221 | Chr20 |
| | Scaffold_22 | 75,693,831 | Chr21 |
| | Scaffold_23 | 72,542,000 | Chr22 |
| | Scaffold_24 | 66,255,986 | Chr23 |
| | Scaffold_25 | 64,861,969 | Chr24 |
| | Scaffold_26 | 51,220,874 | Chr25 |
| | Scaffold_27 | 50,201,775 | Chr26 |
| | Scaffold_28 | 38,686,079 | Chr27 |
| Unplaced scaffold | Scaffold_29 | 910,287 | unplaced_scaffold_1 |
| | Scaffold_30 | 647,422 | unplaced_scaffold_2 |
| | Scaffold_31 | 402,282 | unplaced_scaffold_3 |
| | Scaffold_32 | 226,713 | unplaced_scaffold_4 |
| | Scaffold_33 | 158,269 | unplaced_scaffold_5 |
| | Scaffold_34 | 102,165 | unplaced_scaffold_6 |
| | Scaffold_35 | 85,603 | unplaced_scaffold_7 |
| | Scaffold_36 | 64,000 | unplaced_scaffold_8 |
| | Scaffold_37 | 63,279 | unplaced_scaffold_9 |
| | Scaffold_38 | 53,000 | unplaced_scaffold_10 |
| | Scaffold_39 | 36,223 | unplaced_scaffold_11 |

**Table S3. Coverage depth for sequenced reads, related to Table 1.**

| Coverage depth for HiFi reads | |
|---|---|
| **Depth (X)** | **Coverage ratio (%)** |
| 1 | 100% |
| 4 | 99.99% |
| 10 | 99.99% |
| 20 | 99.81% |
| Coverage depth (×) | 44.5514 |
| Mapping rate (%) | 100% |

| Coverage depth for NGS reads | |
|---|---|
| **Depth (X)** | **Coverage ratio (%)** |
| 1 | 99.85% |
| 5 | 99.32% |
| 10 | 97.87% |
| 20 | 94.07% |
| Coverage depth (×) | 50.0252 |
| Mapping rate (%) | 99.91% |

**Table S4. Assessment of the African manatee genome completeness using BUSCO, related to Table 1**. Completeness evaluation of the assembled genome was based on "mammalia_odb10" dataset in BUSCO software.

| Type | Number | Percent (%) |
|---|---|---|
| Complete BUSCOs (C) | 8,885 | 96.3 |
| Complete and single-copy BUSCOs (S) | 8,801 | 95.4 |
| Complete and duplicated BUSCOs (D) | 84 | 0.9 |
| Fragmented BUSCOs (F) | 81 | 0.9 |
| Missing BUSCOs (M) | 260 | 2.8 |
| Total BUSCO groups searched | 9,226 | 100 |

**Table S5. Repeat content of the African manatee genome estimated by RepeatMasker software, related to Figure 1.**

| Type | Super family | Number of elements | Length of sequence (bp) | Percentage of sequence (%) |
|---|---|---|---|---|
| SINEs | | 1,998,302 | 311,262,245 | 10 |
| | Alu/B1 | 7 | 396 | 0 |
| | MIRs | 562,887 | 76,371,731 | 2 |
| LINEs | | 1,678,457 | 1,028,482,630 | 32 |
| | LINE1 | 937,452 | 715,751,952 | 22 |
| | LINE2 | 426,216 | 103,210,041 | 3 |
| | L3/CR1 | 55,305 | 11,775,216 | 0 |
| | RTE | 258,238 | 197,510,326 | 6 |
| LTR elements | | 628,166 | 274,344,453 | 9 |
| | ERVL | 148,094 | 88,340,305 | 3 |
| | ERVL-MaLRs | 324,307 | 144,095,618 | 5 |
| | ERV_classI | 50,797 | 25,491,630 | 1 |
| | ERV_classII | 73,722 | 7,859,498 | 0 |
| DNA elements | | 452,822 | 94,276,130 | 3 |
| | hAT-Charlie | 234,163 | 94,276,130 | 1 |
| | TcMar-Tigger | 86,691 | 23,488,891 | 1 |
| Unclassified | | 11,415 | 1,973,071 | 0 |
| Total interspersed repeats | | | 1,710,338,529 | 54 |
| Small RNA | | 48 | 4,773 | 0 |
| Satellites | | 177,862 | 103,822,642 | 3 |
| Simple repeats | | 0 | 0 | 0 |
| Low complexity | | 0 | 0 | 0 |
| Total repeats | | | 1,814,613,450 | 57 |

**Table S6. Repeat sequence statistics of the African manatee genome integrated by four software, related to Figure 1.**

| Software | Repeat Size (bp) | Percentage of sequence (%) |
|---|---|---|
| TRF | 48,625,231 | 1.526 |
| RepeatMasker | 1,814,609,246 | 56.962 |
| RepeatProteinMask | 632,041,885 | 19.84 |
| RepeatModeler | 1,731,632,266 | 54.357 |
| Total | 1,965,196,057 | 61.689 |

**Table S7. Basic statistics of the protein-coding genes for the African manatee and related species, related to Figure 1.**

| Scientific name | English name | Refseq accession | Total number of genes | Average gene Length (bp) | Average CDS length (bp) | Average CDS number per gene | Average CDS length (bp) | Average intron length (bp) |
|---|---|---|---|---|---|---|---|---|
| *Trichechus senegalensis* | African manatee | This study | 20,590 | 52,066.52 | 1,678.93 | 9.69 | 173.31 | 5,799.93 |
| *Bos taurus* | Cattle | GCF_002263795.3 | 20,976 | 53,243.84 | 1,678.34 | 9.46 | 177.40 | 5,001.99 |
| *Choloepus didactylus* | Linnaeus's two-toed sloth | GCF_015220235.1 | 23,536 | 54,557.99 | 1,593.41 | 8.54 | 186.59 | 5,805.38 |
| *Canis lupus familiaris* | Dog | GCF_011100685.1 | 20,950 | 52,657.04 | 1,683.14 | 9.54 | 176.49 | 4,795.65 |
| *Elephas maximus* | Asian elephant | GCF_024166365.1 | 21,809 | 60,528.80 | 1,670.96 | 9.05 | 184.63 | 5,893.54 |
| *Homo sapiens* | Human | GCF_009914755.1 | 20,042 | 61,830.58 | 1,735.43 | 9.84 | 176.38 | 5,457.10 |
| *Loxodonta africana* | African savanna elephant | GCF_000001905.1 | 21,081 | 51,694.83 | 1,639.09 | 9.10 | 180.13 | 5,366.00 |
| *Trichechus manatus* | West Indian manatee | GCF_000243295.1 | 19,095 | 57,826.51 | 1,712.99 | 9.88 | 173.43 | 5,603.70 |

**Table S8. Assessment of the African manatee genome annotation completeness using BUSCO, related to Table 1.** Completeness evaluation of the assembled genome annotation was based on "mammalia_odb10" dataset in BUSCO software.

| Type | Number | Percent (%) |
|---|---|---|
| Complete BUSCOs (C) | 9,040 | 98 |
| Complete and single-copy BUSCOs (S) | 8,978 | 97.3 |
| Complete and duplicated BUSCOs (D) | 62 | 0.7 |
| Fragmented BUSCOs (F) | 5 | 0.1 |
| Missing BUSCOs (M) | 181 | 1.9 |
| Total BUSCO groups searched | 9,226 | 100 |

**Table S18. The IUCN status and genome-wide heterozygosity (π) of three Sirenia species and 25 other mammalian species, related to Figure 4.**

| Common name | Species name | IUCN status | Observed pi | Source | DOI |
|---|---|---|---|---|---|
| Baiji | *Lipotes vexillifer* | CR | 0.000121 | Zhou et al., 2013 | 10.1038/ncomms3708 |
| Brown Hyena | *Parahyaena brunnea* | NT | 0.000121 | Westbury et al., 2018 | 10.1093/molbev/msy037 |
| Narwhal | *Monodon monoceros* | LC | 0.00014 | Westbury et al., 2019 | 10.1016/j.isci.2019.03.023 |
| Iberian lynx | *Lynx pardinus* | EN | 0.00018 | Westbury et al., 2018 | 10.1093/molbev/msy037 |
| Cheetah | *Acinonyx jubatus* | VU | 0.0002 | Dobrynin et al., 2015 | 10.1186/s13059-015-0837-4 |
| Killer whale | *Orcinus orca* | DD | 0.00021 | Westbury et al., 2018 | 10.1093/molbev/msy037 |
| Snow leopard | *Panthera uncia* | VU | 0.00023 | Cho et al., 2013 | 10.1038/ncomms3433 |
| Beluga whale | *Delphinapterus leucas* | LC | 0.00029 | Westbury et al., 2019 | 10.1016/j.isci.2019.03.023 |
| Tasmanian devil | *Sarcophilus harrisii* | EN | 0.00032 | Miller et al., 2011 | 10.1073/pnas.1102838108 |
| Polar bear | *Ursus maritimus* | VU | 0.00032 | Westbury et al., 2018 | 10.1093/molbev/msy037 |
| Amur tiger | *Panthera tigris altaica* | EN | 0.000486 | Cho et al., 2013 | 10.1038/ncomms3433 |
| African manatee | *Trichechus senegalensis* | VU | 0.000497 | This study | This study |
| African lion | *Panthera leo* | VU | 0.000583 | Cho et al., 2013 | 10.1038/ncomms3433 |
| Minke whale | *Balaenoptera acutorostrata* | LC | 0.00061 | Yim et al., 2014 | 10.1038/ng.2835 |
| Eastern lowland gorilla | *Gorilla beringei graueri* | CR | 0.00064 | Xue et al., 2015 | 10.1126/science.aaa3952 |
| Chinese pangolin | *Manis pentadactyla* | CR | 0.00085 | Heighton et al., 2023 | 10.1093/molbev/msad190 |
| Indo-Pacific finless porpoise | *Neophocaena phocaenoides* | VU | 0.00093 | Morin et al., 2020 | 10.1111/1755-0998.13284 |
| Indo-Pacific bottlenose dolphin | *Tursiops aduncus* | NT | 0.00095 | Vijay et al., 2018 | 10.1093/molbev/msy108 |

| | | | | | |
|---|---|---|---|---|---|
| Yangtze finless porpoise | *Neophocaena asiaeorientalis asiaeorientalis* | EN | 0.00105 | Morin et al., 2020 | 10.1111/1755-0998.13284 |
| Florida manatee | *Trichechus manatus latirostris* | VU | 0.00106 | This study | This study |
| Giant panda | *Ailuropoda melanoleuca* | VU | 0.00135 | Leffler et al., 2012 | 10.1371/journal.pbio.1001388 |
| Bottlenose dolphin | *Tursiops truncatus* | LC | 0.00142 | Yim et al., 2014 | 10.1038/ng.2835 |
| Fin whale | *Balaenoptera physalus* | VU | 0.00151 | Yim et al., 2014 | 10.1038/ng.2835 |
| Dugong | *Dugong dugon* | VU | 0.00166 | This study | This study |
| Olive baboon | *Papio anubis* | LC | 0.00189 | Corbett-Detig_2015 | 10.1371/journal.pbio.1002112 |
| Blue whale | *Balaenoptera musculus* | EN | 0.0021 | Morin et al., 2020 | 10.1111/1755-0998.13284 |
| Sperm whale | *Physeter catodon* | VU | 0.00228 | Morin et al., 2020 | 10.1111/1755-0998.13284 |
| Przewalski's horse | *Equus ferus przewalskii* | EN | 0.00363 | Corbett-Detig et al., 2015 | 10.1371/journal.pbio.1002112 |

**Table S19. The mitochondrial genome accession for phylogeny reconstruction of the order Sirenia, related to STAR Methods.**

| Scientific name | English name | Accession number |
|---|---|---|
| *Trichechus senegalensis* | African manatee | MW073827.1 |
| *Trichechus senegalensis* | African manatee | This study |
| *Trichechus manatus* | West Indian manatee | NC_010302.1 |
| *Trichechus inunguis* | Amazonian manatee | MW073826.1 |
| *Dugong dugon* | Dugong | NC_003314.1 |
| *Loxodonta africana* | African elephant | NC_000934.1 |
| *Elephas maximus* | Asian elephant | NC_005129.2 |

**Table S20. Genome accession for the mammals used in genome-wide comparative genomic analysis, related to STAR Methods.**

| Order | Scientific name | English name | Abbr. | GenBank assembly accession | RefSeq assembly accession |
|---|---|---|---|---|---|
| Artiodactyla | *Balaenoptera musculus* | Blue whale | Bmus | GCA_009873245.2 | GCF_009873245.2 |
| Artiodactyla | *Bos taurus* | Cattle | Btau | GCA_002263795.4 | GCF_002263795.3 |
| Artiodactyla | *Camelus ferus* | Wild Bactrian camel | Cfer | GCA_009834535.1 | GCF_009834535.1 |
| Artiodactyla | *Capra hircus* | Goat | Chir | GCA_001704415.2 | GCF_001704415.2 |
| Artiodactyla | *Sus scrofa* | Wild boar | Sscr | GCA_000003025.6 | GCF_000003025.6 |
| Artiodactyla | *Tursiops truncatus* | Common bottlenose dolphin | Ttru | GCA_011762595.1 | GCF_011762595.1 |
| Carnivora | *Canis lupus familiaris* | Dog | Clup | GCA_014441545.1 | GCF_014441545.1 |
| Carnivora | *Felis catus* | Cat | Fcat | GCA_018350175.1 | GCF_018350175.1 |
| Chiroptera | *Phyllostomus discolor* | Pale spear-nosed bat | Pdis | GCA_004126475.2 | GCF_004126475.2 |
| Chiroptera | *Rhinolophus ferrumequinum* | Greater horseshoe bat | Rfer | GCA_004115265.2 | GCF_004115265.2 |
| Dasyuromorphia | *Sarcophilus harrisii* | Tasmanian devil | Shar | GCA_902635505.1 | GCF_902635505.1 |
| Perissodactyla | *Equus caballus* | Horse | Ecab | GCA_002863925.1 | GCF_002863925.1 |
| Pilosa | *Choloepus didactylus* | Linnaeus's two-toed sloth | Cdid | GCA_015220235.1 | GCF_015220235.1 |
| Primates | *Homo sapiens* | Human | Hsap | GCA_009914755.1 | GCF_009914755.1 |
| Primates | *Macaca mulatta* | Rhesus macaque | Mmul | GCA_003339765.1 | GCF_003339765.1 |
| Proboscidea | *Elephas maximus* | Asian elephant | Emax | GCA_024166365.1 | GCF_024166365.1 |
| Rodentia | *Rattus norvegicus* | Brown rat | Rnor | GCA_015227675.2 | GCF_015227675.2 |
| Rodentia | *Mus musculus* | House mouse | Mmus | GCA_000001635.27 | GCF_000001635.27 |
| Sirenia | *Trichechus senegalensis* | African manatee | Tsen | This study | This study |

**Table S21. Fossil-calibrated time points for calculating divergence time when reconstructing the phylogeny of the class Mammalia, related to STAR Methods.**

| Node | Node minimum age (Ma) | Node maximum age (Ma) |
|---|---|---|
| *Felis catus* - *Canis lupus familiaris* split | 52.9 | 57.3 |
| *Balaenoptera musculus* - *Tursiops truncatus* split | 32.3 | 35.2 |
| *Bos taurus* - *Capra hircus* split | 19.5 | 30.9 |
| *Phyllostomus discolor* - *Rhinolophus ferrumequinum* split | 57.9 | 64 |
| *Rattus norvegicus* - *Mus musculus split* | 14 | 19 |
| *Homo sapiens* - *Macaca mulatta* split | 27 | 30.6 |
| *Elephas maximus* - *Trichechus senegalensis* split | 56.9 | 64.2 |
| *Homo sapiens* - *Sarcophilus harrisii* split | 147.1 | 164.3 |

**Table S22. The reference genome and SRA accession number for PSMC analysis, related to STAR Methods.**

| Scientific name | GenBank assembly accession | SRA acccesion number | Layout | Platform |
|---|---|---|---|---|
| *Trichechus manatus latirostris* | GCA_000243295.1 | SRR307134<br>SRR307135<br>SRR307136<br>SRR307137<br>SRR314670<br>SRR328413<br>SRR328414<br>SRR328415<br>SRR328416<br>SRR328417<br>SRR331132<br>SRR331133<br>SRR331135<br>SRR331146<br>SRR331136<br>SRR331138<br>SRR331140<br>SRR331141<br>SRR331143<br>SRR331144<br>SRR331145<br>SRR331147<br>SRR331148 | PAIRED | Illumina HiSeq 2000 |
| *Dugong dugon* | GCA_905400935.1 | SRR17854495<br>SRR17870680 | PAIRED | Illumina NovaSeq 6000 |
| *Trichechus senegalensis* | This study | This study | PAIRED | DNBSEQ-T7 |